

Measuring AI Ability to Complete Long Software Tasks

50
%

task-completion time horizon doubling
every ~7 months

Time Horizon Doubling Every 7 Months

Thomas et al. — Becker, et al. — METR

arxiv.org/abs/2503.14499 (March 2025)

AI
Progress

2×

every
7 mo

		Length	Description
task definition	1 minute	1 minute	Multiple choice: "Which file is a text script?" (Correct: "text.py", "text.sh", "text.txt")
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New Metric: 50% Task-Completion Time Horizon

The 50% task-completion time horizon = duration of tasks (measured by skilled human professionals) that an AI model can complete with 50% success rate

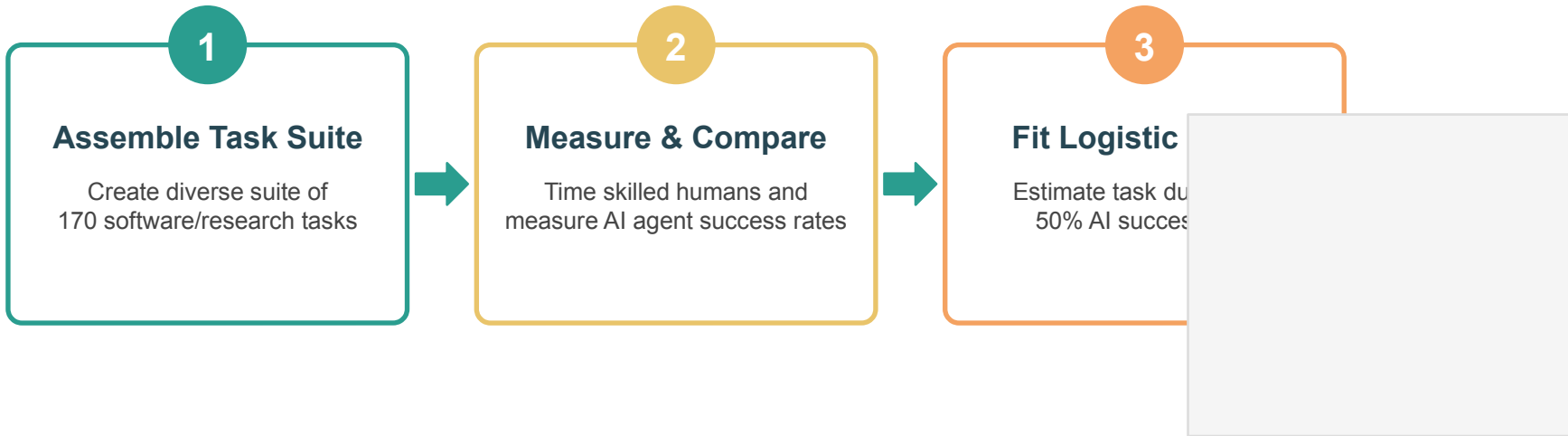


Fig. 2: Methodology illustration

Key Insight: This metric allows direct comparison of AI capabilities across different task durations and complexity levels.

170 Tasks Spanning 3 Seconds to 8 Hours of Human Effort

Over 800 human baselines totaling 2,529 hours | Task durations range from 3 seconds to 8 hours

Family	Tasks	Example Task	Duration
HCAST	97	oxdna_simple (detect/fix bug in molecular dynamics)	9 min
RE-Bench	7	cuda_backtesting (CUDA kernels for backtesting)	8 hours
SWAA	66	find_shell_script (which file is a shell script)	3 sec

170

Total Tasks

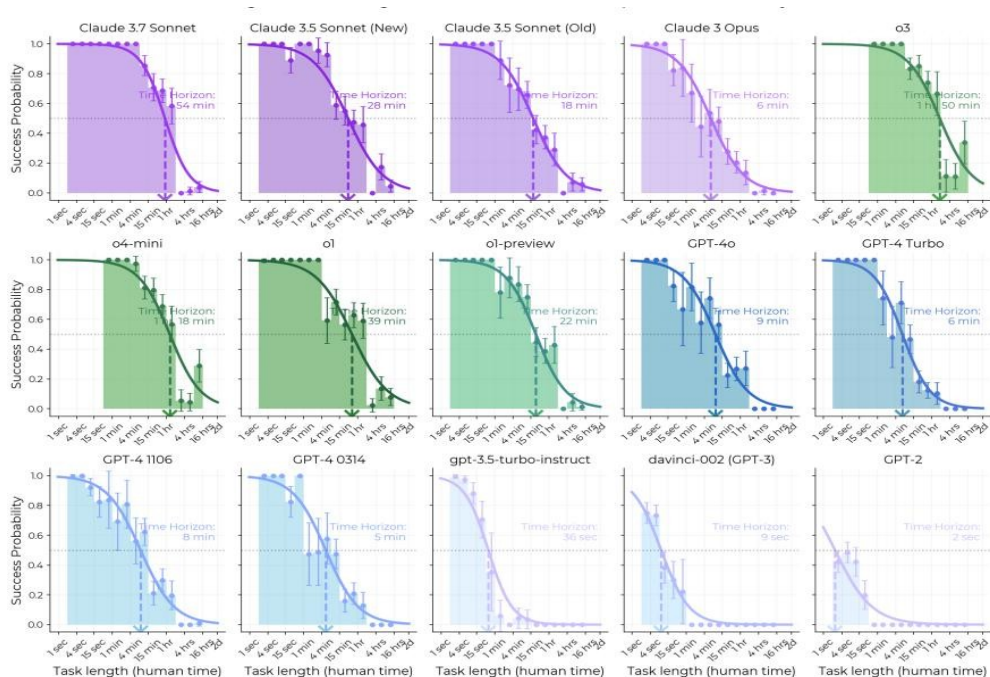
3 sec - 8 hrs

Duration Range

800+ / 2,529 hrs

Human Baselines

AI Task Horizon Grew from 2 Seconds (GPT-2) to ~1 Hour 50 Minutes (o3) in 6 Years



6 orders of magnitude in 6 years (seconds → hours)

KEY FINDINGS

Doubling Time

~207 Days

(~7 months)

$R^2 = 0.97$

(excellent fit)

2023–2025 growth ~20% faster

GPT-2

~2 sec

GPT-4

~5 min

o3

~1h 50m

Model	Year	50% Horizon
GPT-2	2019	~2 sec
GPT-4 (0314)	2023	~5 min
Claude 3.7 Sonnet	2025	~54 min
o3	2025	~1h 50m

Success Probability Drops Sharply as Task Length Increases for All Models

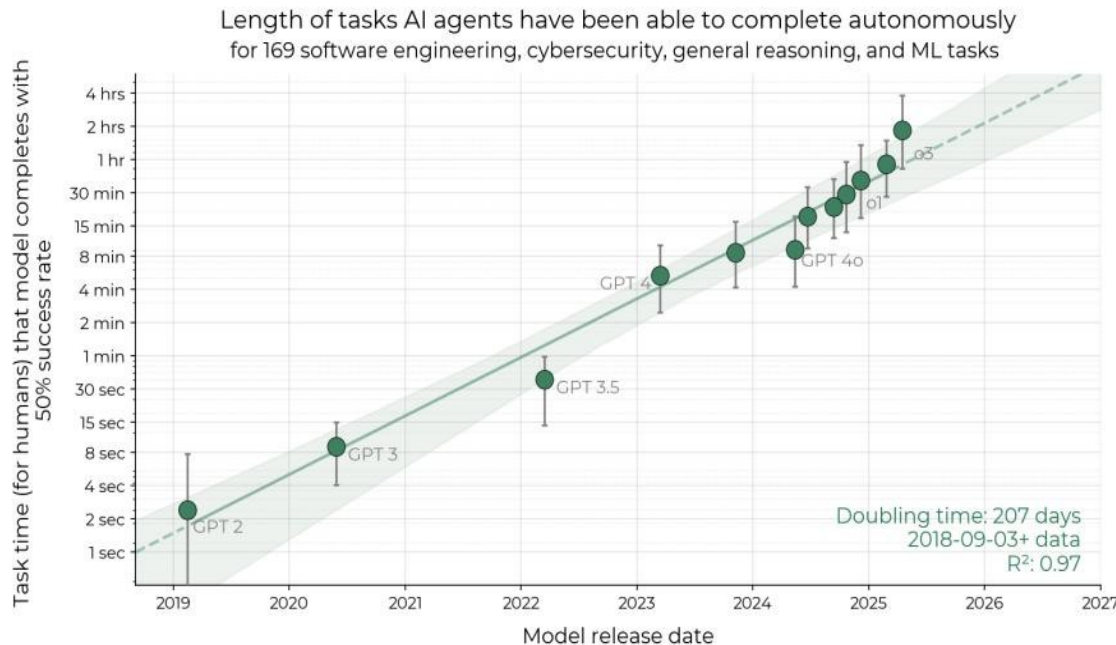


Figure 3: Per-model success probability vs task duration

GPT-2 (2 sec)

Claude 3.7 Sonnet (54 min)

o3 (1h 50min)

Frontier models extend time horizon significantly

Key Findings

- Success probability curves show sharp decline as task duration increases for all models
- Frontier models (Claude 3.7 Sonnet 54min, o3 1h50m) push curves far rightward vs GPT-2 (2 sec)
- 80% time horizon is ~5x shorter than 50% time horizon across all models
- All models show same pattern: higher success on shorter tasks, sharply declining as duration increases

Reasoning, Tool Use, and Error Recovery Drive Time Horizon Growth

KEY DRIVERS



Improved Logical Reasoning

Models can plan and execute multi-step solutions



Better Tool Use

Models effectively use computers, APIs, code interpreters



Greater Reliability

Models recognize errors and self-correct



Ability to Adapt

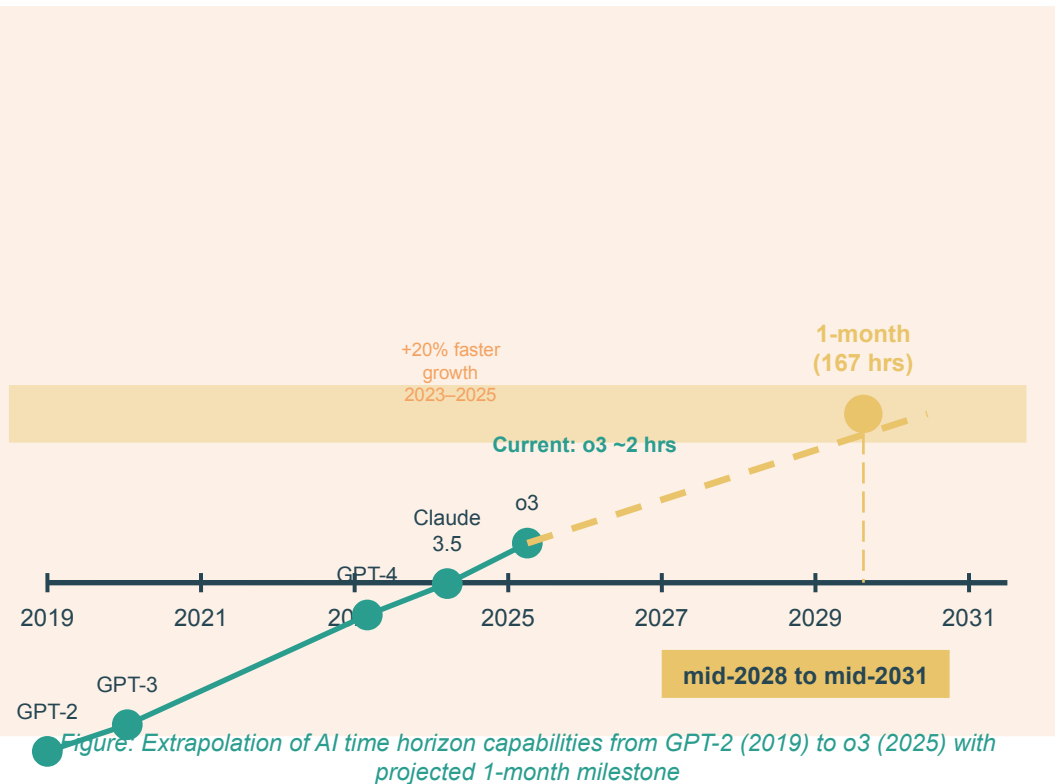
Models learn from failures and try alternative approaches

KEY LIMITATION



Performance is notably lower on less structured, "messier" tasks
External validity remains an open question - tasks do not perfectly represent average real-world intellectual labor

Extrapolation Predicts Month-Long Task Automation by 2028–2031



Extrapolation

If the exponential trend continues, AI reaches >1 month time horizon (167 work hours) between mid-2028 and mid-2031 — implying automation of many month-long software projects.

Key Caveats

- External validity remains an open question
- Tasks may not perfectly represent real-world intellectual labor
- Trend could slow down or speed up
- Supplementary experiments found little evidence trends are slower on more realistic tasks

Takeaways: Exponential Autonomous Capability Growth Demands Proactive Safety Governance

1 Robust Metric

50% time horizon is a robust, intuitive metric for tracking AI autonomy with $R^2=0.97$ over 6 years of

2 Doubling Every ~7 Months

Doubling time is ~7 months with possible acceleration since 2024

3 Current Frontier

Current frontier AI agents (o3) can autonomously complete tasks that take skilled humans nearly 2 hours

4 External Validity

External validity remains an open question – tasks may not perfectly represent real-world intellectual labor

5 Key Limitations

Poor performance on unstructured tasks and uncertain generalization to real-world job distributions

The exponential growth in autonomous AI capabilities demands proactive safety governance.